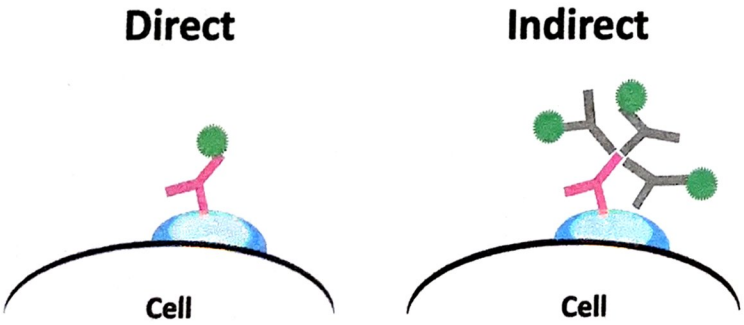
Indirect Immunofluorescence Protocol

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Safety statement: Prior to beginning an experiment, please read the Standard Operating Procedure (SOP) for this protocol as well as the Safety Data Sheets (SDS) for all chemical reagents listed. The SOP will contain a list of all chemicals for this protocol as well as directions for handling and disposal of hazardous wastes. It will also contain proper PPE guidelines and emergency protocols for accidents involving hazardous chemicals. Always be aware of your surroundings in the lab and be sure to communicate any hazards to those around you when performing an experiment.

Hazard statement: Any hazardous chemicals handled within this protocol will be marked with the following notation: (H). Please refer to the associated SOP to better understand the hazards and how to work with and dispose of these chemicals safely.

Background: Immunofluorescence allows a scientist to observe the physical locations of certain proteins through binding of specific antibodies to the protein(s) of interest. Similar to a Western blot where chemiluminescent signal is amplified through a secondary antibody cross-adsorbed to an enzyme that produces light, "AlexaFluor" secondary antibodies are conjugated to a fluorophore that is excited at a specific wavelength of light. This allows simultaneous labeling of multiple proteins and consequently the ability to observe co-localization of two proteins. This is a semi-quantitative method of measuring protein-protein interactions because the fluorescence intensity can be measured, but antibody-antigen interactions are not 100% reliable, so the quality of the antibody being used plays a large factor. Oftentimes, this method is supplemented by another unique approach that measures the same biological phenomenon. This technique can also be combined with other techniques to answer diverse biological questions.



Equipment:

1. 6-well plates
2. 18x18mm glass coverslips
3. Glass microscope slides
4. Light microscope with proper filter cubes or confocal microscope with appropriate excitation wavelength lasers

Reagents:

5. 1X phosphate-buffered saline (PBS)
6. Bovine serum albumin (BSA)
7. 10% Triton X-100
8. 37% formaldehyde
9. Primary antibodies
10. Secondary antibodies (i.e. AlexaFluors)
11. 4',6-diamidino-2-phenylindole (DAPI)
12. Mountant
13. Clear nail polish

Procedural Overview: In order to label proteins within an intact cell, the cellular structures and their integrities must be maintained, or "fixed." This is done with a crosslinking agent called formaldehyde. It is a zero-length crosslinker that covalently cross-links any organic molecules that contain primary amine groups (-NH2). Once fixed, the cell membranes are permeabilized with a detergent like Triton X-100, which disrupts lipid packing in the membrane bilayer, thus creating small pores in the membrane. This allows larger molecules like antibodies to enter the fixed cells and bind to their targets. Prior to incubating cells with antibodies, they are washed in a blocking solution made up of bovine serum albumin (BSA). This protein binds all nonspecific binding sites in the cells to increase binding between the antibody and its target. After this, the cells are incubated in a primary antibody solution diluted in block buffer. The primary antibodies will bind specific regions of your protein of interest. Following a few washes, the cells are incubated with secondary antibodies cross-adsorbed to "AlexaFluors," small fluorophores that are excited with specific wavelengths of light. Generally, the nuclei of the cells are also stained with intercalating agents like 4',6-diamidino-2-phenylindole (DAPI) or Hoechst stain. These agents absorb ultraviolet light and emit in the blue region of the visible spectrum. Finally, the coverslips with fluorescently-labeled cells must be affixed to a microscope slide in order to visualize the sample using a microscope. This is done with mountant, a substance that adheres the coverslip to the slide and provides a clear refractive index for imaging. Mountants often contain antifade solutions, which prevent the fluorophores from photobleaching, or "fading," as it's seen on the microscope. In order to prevent the coverslips from sliding on the microscope with the immersion oil, the coverslips are secured to the slide with something like nail polish or rubber cement applied to the edges of the coverslips.

Procedures:

1. Seed cells at ~70% confluence on glass coverslips and grow overnight at 37°C and 5% CO_2.
    a. If cells do not attach well to the glass, consider coating coverslips with poly-L-lysine prior to seeding.
2. Next day, remove medium and gently wash cells twice with 1X PBS on a rocker for 5 min per wash.
3. Prepare 3.7% formaldehyde diluted in 1X PBS. Incubate coverslips for 15 min at room temperature with gentle rocking to fix the cells.
4. Remove the formaldehyde and wash coverslips 3X with 1X PBS on a rocker for 5 min per wash.
5. Prepare 0.2% Triton X-100 diluted in 1X PBS. Incubate coverslips for 25 min at room temperature with gentle rocking to permeabilize the cell membranes.
6. Remove the detergent and wash coverslips 3X with 1X PBS on a rocker for 5 min per wash.
7. Prepare 3% BSA + 0.1% Triton X-100 in 1X PBS. Incubate for 1 hour at room temperature OR overnight at 4°C with gentle rocking.
    a. BSA is lyophilized. Prepare the 3% solution as weight-by-volume (e.g. 3% of 50mL total volume would be 1.5g BSA).
8. Wash the coverslips 3X with 1X PBS, 5 min per wash.
9. Prepare the primary antibody solution by diluting it to the appropriate concentration in block buffer.
10. Using plastic Tupperware and parafilm, pipette 50µL of primary antibody solution onto the parafilm as a bead of liquid. Using tweezers, take each coverslip and wick off excess liquid using a KimWipe. Carefully lower the coverslips **cell-side down** onto the bead of liquid. Incubate for 1 hour at room temperature OR overnight at 4°C.
11. Remove coverslips and place back into 6-well plate with 1X PBS. Wash coverslips 3X for 5 min each.
12. Prepare the secondary antibody solution the same way as the primary antibody. All AlexaFluor secondary antibodies are diluted 1:200.
13. Lower coverslips onto parafilm with antibody solutions and incubate for 1 hour at room temperature. ****GOING FORWARD YOU MUST PROTECT FROM LIGHT****
14. Remove coverslips and place back into 6-well plate with 1X PBS. Wash coverslips 3X for 5 min each **wrapped in foil.**
15. Dilute DAPI (1mg/mL stock) 1:4,000 in 1X PBS and incubate with coverslips for 5 min at room temperature with gentle rocking.
16. Remove DAPI and wash coverslips 3X with 1X PBS.
17. After third wash, use tweezers to wick off excess liquid from the coverslips before mounting them cell-side down on microscope slides.
18. Mount for 24 hours overnight.
19. Next day, add nail polish to edges of the coverslip to seal them. Let dry 15 minutes before imaging. See "Confocal Microscopy Protocol."